

Publication Roadmap

S&P 500 Stock Outperformer Classification Using Financial Ratios

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Inspiration: Ananthakumar & Sarkar (2017), IEEE DASC/PiCom/DataCom

What the 2017 Paper Did — and Why Replication Alone Is Not Enough

Ananthakumar & Sarkar (2017) applied basic logistic regression to annual S&P 500 data and demonstrated that financial ratios could distinguish outperformers from underperformers. This was a meaningful contribution in 2017. However, simply replicating the same method on updated data in 2025 does not constitute a novel academic contribution and would not pass peer review.

A publishable paper must answer a question that has not been answered before, or answer an old question with meaningfully better data, methods, or scope.

Core Research Question (Your Novel Angle)

Can quarterly financial ratios — combined with rigorous class-imbalance correction, multicollinearity screening (VIF), and comparison across multiple classification models — reliably predict S&P 500 stock outperformance across different market regimes (pre-COVID, COVID crisis, post-COVID recovery)?

This question is novel because: (1) the 2017 paper used annual data — quarterly data has not been studied for this problem; (2) no prior study has tested this framework across the COVID regime break; (3) multi-model comparison with proper class-imbalance handling has not been applied to this specific outperformer classification task.

Publication Roadmap — Novel Contributions and Data Requirements

1	Temporal / Regime Analysis or Publishable Research Angles (Choose One or Combine) Train on pre-COVID (2015–2019), test on COVID crash (2020), retest on recovery (2021–2023). Ask: does the model's predictive power break down during market crises? This is practically important and not covered in the literature.
2	Quarterly vs Annual Data Comparison The 2017 paper used annual ratios. Compare model accuracy when trained on quarterly vs annual data. Ask: does higher frequency financial data improve outperformer prediction? No study has made this comparison for S&P 500 classification.
3	Sector-Specific Models Build one logistic regression model per GICS sector (Technology, Healthcare, Financials, etc.). Ask: do different financial ratios matter in different industries? A P/E ratio means something very different for a tech firm vs a utility company.
4	Multi-Model Comparison with Portfolio Backtesting ← Most Publishable Compare LR, LDA, Random Forest, and XGBoost on the same dataset. Apply SMOTE for class imbalance. Report AUC, F1, and simulate an actual portfolio of the top predicted outperformers each quarter and measure returns vs SPY benchmark.

Data Requirements — Honest Assessment

Current Data Source	Problem	Fix Required	Priority
Yahoo Finance (annual, 110 stocks)	Incomplete ratios, small sample, no quarterly	Supplement with SEC EDGAR quarterly	Done
SEC EDGAR quarterly (108 stocks, 3,069 rows)	Missing quarterly stock price/return data	Add quarterly returns via yfinance	Next step
No WRDS / Compustat	Reviewers expect gold-standard data — 2 —	Register at WRDS via Binghamton University	Important

Publication Roadmap — Deliverables and Next Steps

	What Can Be Built for This Project Full Data Pipeline Merge quarterly SEC EDGAR financials + quarterly stock returns (yfinance) into one clean, analysis-ready dataset. Add WRDS/Compustat once access is granted.
	Multicollinearity Screening (VIF) Calculate Variance Inflation Factors for all financial ratios. Remove or combine variables with $VIF > 10$ before modeling to ensure stable logistic regression coefficients.
	Class Imbalance Correction (SMOTE) Apply Synthetic Minority Oversampling Technique so the model learns equally from outperformers and underperformers— prevents the model from always predicting the majority class.
	Four Competing Models Train Logistic Regression, Linear Discriminant Analysis, Random Forest, and XGBoost on identical data. Report Accuracy, AUC-ROC, Precision, Recall, and F1 for each.
	Portfolio Backtesting Simulate buying the top-N predicted outperformers at the start of each quarter. Track portfolio return vs SPY benchmark. Compute Sharpe ratio and max drawdown.
	Publication-Quality Report Academic PDF covering: Abstract, Literature Review, Methodology, Results, Discussion, Conclusion— in the format expected by IEEE or a finance journal.

	Recommended Next Steps — In Order	
1	Register for WRDS Go to wrds-web.wharton.upenn.edu with your @binghamton.edu email. Request access to Compustat. Approval takes 1-3 days.	WRDS
2	Decide on research angle Choose which of the four novel angles to pursue (Regime analysis recommended as most impactful).	Decision
3	Merge quarterly returns Add quarterly stock return data to the existing SEC EDGAR dataset to create the outperformer label per quarter.	Data
	Begin full analysis VIF screening → SMOTE → model training → backtesting → paper.	